# PortableTAK Setup and Troubleshooting Guide

A complete TAK server (OpenTAKServer) that lives on a USB drive and runs on an OpenMANET node or a Windows computer.

## 1. What PortableTAK is

The drive holds the whole server: its software, its database, its users and certificates, and its settings. The computer or node it is plugged into only provides power, network and processing. Move the drive and the same server, with the same users and phones, comes up somewhere else.

- **One drive = one server.** Each drive has its own certificate authority, users and data.
- **Runs on:** OpenMANET nodes (Raspberry Pi), Windows 11 computers (Intel/AMD and ARM) and Apple Silicon Macs. The drive carries software for both processor types and picks the right one automatically.
- **Moves between hosts:** on the first start on a new node or computer, the server adapts itself to that host (addresses and server certificate). Phones already set up keep working.
- **Works offline.** Internet is only needed for optional Remote Access (Tailscale).

### What is on the drive

Plugged into Windows or a Mac, only a small **PORTABLETAK** volume is visible. It holds:

| | |
|---|---|
| **Start PortableTAK (Windows)** / **(Mac)** | Runs the server on this computer. |
| **Scan to open the dashboard (node).png** | QR code for the dashboard when the drive is in a node. |
| **Quick Start**, **Quick Reference**, this guide | Documentation (PDF). |

The server itself is on a second, Linux partition that Windows and macOS cannot read. If Windows offers to **format** or macOS offers to **initialize** a disk, always choose **Cancel** or **Ignore**.

## 2. Making a drive (PortableTAK Setup)

Drives are made with **PortableTAK Setup**, on a Windows 11 computer or a Mac (Apple Silicon). Both make the same drive.

### On Windows 11

1. Unzip **PortableTAK-Windows-Setup**. Plug in a USB drive of **16 GB or more** (it will be erased).
2. Double-click **PortableTAK Setup (Windows)** and click **Yes**. The first time, it may install WSL and ask you to restart; then run it again.
3. Choose a ready-to-use drive (name and passwords now) or a blank drive, pick the USB drive from the list and type the confirmation phrase.
4. Wait about 10 minutes. The drive is built, written and verified.

### On a Mac

1. One-time: install Homebrew, then in Terminal: `brew install colima docker`.
2. Plug in a USB drive of **16 GB or more**. It will be erased.

3. Double-click **PortableTAK Setup.command** (first time: right-click → Open).
4. Enter the **server name**, the **administrator password** (dashboard sign-in) and the **client password** (protects certificate packages; used for QR enrollment). Passwords: 8+ characters, 12+ recommended, and the two must be different. Typos can be retried.
5. Pick the drive from the list and type `ERASE diskN` exactly as shown. macOS asks for your Mac password.
6. Wait about 8–10 minutes: setup, writing and verification. The drive is ejected when done.

**Blank drives:** choose option 2 at the start to make a drive that is not set up yet (useful for preparing several drives). The first time a blank drive starts on Windows, **Start PortableTAK** asks for the server name and both passwords and sets it up on that computer. A blank drive cannot be set up from a node.

> Setup uses the Mac's internet once (map icon sets). Without internet the drive still works; the icons download later the first time the server runs with internet.

> **Never clone a set-up drive** to make more. Clones share the same keys, users and certificate authority. Always make each drive with PortableTAK Setup.

## 3. Using the drive on an OpenMANET node

The node needs the **PortableTAK loader** (part of the field extension, or installed once by an administrator with `portabletak-loader-install.sh`). With it, the first time a PortableTAK drive is plugged in the node installs PortableTAK from the drive by itself (about 5 minutes extra), and a drive made with a newer release updates the node the same way. That includes a drive whose Start Here files were refreshed from a newer release on a Mac or PC: the node picks up the newer package from the Start Here partition. Only packages signed with the PortableTAK release key are accepted.

1. Plug the drive into a **blue USB 3 port** on a running node. It starts by itself in 3–5 minutes, and also after every reboot or power loss.
2. Connect a phone or laptop to the node's Wi-Fi and open `https://10.41.254.1:8444` (or scan the QR code on the drive). Accept the certificate warning (Advanced → Continue).
3. Sign in as `administrator`.

**Addresses on a node:** phones on the node's own Wi-Fi use `10.41.254.1`. Devices elsewhere on the mesh use `10.41.250.10`. Both are in every client package.

**Internet sharing:** if the node's Ethernet port has internet, the node shares it with its Wi-Fi clients automatically and stops when the uplink goes away (phones then use cellular for internet).

**Unplugging:** press **Safe Remove** in the dashboard and wait for "safe to remove". Pulling the drive or power without this is survivable: the next start checks and repairs the drive. Avoid it when you can.

## 4. Using the drive on a Windows computer

### Requirements

- Windows 11 version 22H2 or newer (Intel/AMD or ARM). Administrator rights (Windows asks).
- Virtualization enabled (it is on most computers; WSL reports it if not).

### Start

1. Plug in the drive, open the **PORTABLETAK** drive in File Explorer, double-click **Start PortableTAK (Windows)**, click **Yes**.
2. First time on a computer only (needs internet): it installs WSL (Windows Subsystem for Linux) if missing (type **Y**; then restart and start it again), installs **usbipd** (Microsoft's recommended tool that hands the USB drive to WSL), turns on WSL mirrored networking, opens the firewall for PortableTAK, and installs the small PortableTAK host. The first start of the server on a computer takes a few minutes.
3. If the drive is blank, the window asks for the server name and passwords (typos can be retried) and sets it up (5–10 minutes).
4. The window shows **this computer's address**. Phones and WinTAK use it. The dashboard opens in the browser.
5. While the server runs, the drive is attached to the PortableTAK host and disappears from File Explorer. Desktop shortcuts appear: **Stop PortableTAK**, **PortableTAK Dashboard**, **PortableTAK Status**.

### Stop

Double-click **Stop PortableTAK** on the desktop. Wait for "You can remove the drive now", then unplug. On a computer the dashboard shows this reminder instead of Start/Stop buttons: only the Start and Stop shortcuts can attach and release the drive.

### WinTAK on the same computer

Create a user for WinTAK and use **WinTAK Setup** in the dashboard. WinTAK connects to the computer's address shown at start.

### Start automatically

Run `%LOCALAPPDATA%\PortableTAK\PortableTAK.ps1 install-helper` in PowerShell once. From then on, plugging the drive into that computer starts PortableTAK and opens the dashboard. `remove-helper` turns it off.

> **Laptop on the move:** PortableTAK uses the computer's address at start. If you change networks (for example from home Wi-Fi to a hotspot), stop and start PortableTAK so it re-binds to the new address. Phones keep their certificates. Only the address they connect to changes, so add the new address in the TAK app or use Remote Access.

### Using the drive on a Mac

Apple Silicon Macs can run the server too, in a small Linux virtual machine. One-time preparation: install Homebrew and run `brew install qemu`.

1. Plug in the drive and double-click **Start PortableTAK (Mac)** on the PORTABLETAK volume. Enter your Mac password when asked (it gives the server access to the drive).
2. Blank drives ask for the name and passwords first. The dashboard then opens at `http://localhost:8090`, and the window shows the Mac's address for phones and WinTAK (other devices reach the dashboard at `https://<that address>:8444`). Allow incoming connections for qemu if macOS asks.

3. The PORTABLETAK volume disappears while the server runs. To stop, press Return in the window or double-click **Stop PortableTAK** on the Desktop; then eject and unplug.

## 5. Adding phones and devices

1. **Users** → **Create User**: one user per person. In each TAK app, give every device its own **callsign**.
2. Under **Manage User / Download Setup Package** choose one:

| Device | Use | Notes |
|---|---|---|
| ATAK (Android) | **ATAK Package** | Import → Local SD. Contains every server address (Wi-Fi, mesh, Tailscale). Wi-Fi and Tailscale entries are enabled. The others are present but off. |
| iTAK (iPhone) | **iTAK Package** + **iTAK Remote Package** | iTAK only uses the first server in a package. Import both for Wi-Fi and cellular. |
| WinTAK | **WinTAK Setup** | Import in WinTAK. |
| ATAK / iTAK by QR | **ATAK QR Setup** / **iTAK QR Setup** | The device must trust the server first. For ATAK, import the **ATAK Trust Package (for QR)** from Client Setup once per phone, then scan. A QR code works for 3 uses. In the QR window, pick the network the phone is on (**Phone is on:** Wi-Fi, LAN or Tailscale) if it is not already selected. **iPhone, once:** open the dashboard in Safari on the iPhone → Client Setup → **iPhone Trust Profile (for QR)** → Allow → Settings → Profile Downloaded → Install; then Settings → General → About → Certificate Trust Settings → switch on the PortableTAK CA. Without this, iTAK's QR sign-in fails with "error connecting to the server". (The iTAK Package needs none of this.) With remote access set up, ATAK QR Setup also adds the Tailscale server; on iPhone pick **Phone is on: Tailscale** in the iTAK QR window for cellular. |

3. The server connection turns green in the app.

## 6. Remote access over cellular (Tailscale)

Remote access lets phones reach the server from anywhere they have internet (cellular or any Wi-Fi), without opening ports on a router. It uses **Tailscale**, a free private network: the server and the phones all sign in to the **same Tailscale account** and can then reach each other. Set it up once per server.

### What you need

- A free Tailscale account (`tailscale.com`). Use one account for the server and all its phones.
- Internet on the server side: the node's Ethernet port, or the computer's normal internet.
- The Tailscale app on every phone (App Store / Google Play).

### Step 1: connect the drive (once)

The drive has its own Tailscale identity. It is the same on an OpenMANET node, a Windows computer or a Mac, so phones set up once keep working wherever the drive runs. The computer does not need the

Tailscale app.

1. Make sure the server has internet: the node's Ethernet port (or a phone tethered by USB), or the computer's normal internet.
2. Dashboard → **Remote Access** → **Enable remote access**.
3. Open the link that appears, sign in to Tailscale and press **Connect** to approve the server.
4. Back in the dashboard press **Finish setup**. The server restarts (about 2 minutes) and shows its remote address (100.x.y.z).

### Step 2: set up each phone

1. Install the **Tailscale** app, sign in with the **same account**, and switch it on (allow the VPN profile when asked).
2. Set up ATAK with **ATAK QR Setup** or the **ATAK Package** (Users → your user). Either one gives the phone two servers: the local one and "*server name* (Tailscale)". On iPhone import the **iTAK Remote Package** (or scan **iTAK QR Setup** with "Phone is on: Tailscale") as a second server; keep the normal iTAK one for local Wi-Fi. Phones set up before Step 1 must be set up again.
3. Test: switch the phone's Wi-Fi off. With Tailscale on, the Tailscale server turns green over cellular.

### Step 3: open the dashboard from anywhere

With remote access set up you can manage the server from anywhere with internet, not only on its Wi-Fi.

1. Find the drive's Tailscale address: dashboard → **Remote Access** ("Remote clients: 100.x.y.z or portabletak-xxxxxxxx.….ts.net"). It is the same on every node and computer the drive runs on. Write it down.
2. On the phone or laptop you will use, install the **Tailscale** app, sign in to the **same account** and switch it **on**.
3. Open a browser at `https://<Tailscale address>:8444`, for example `https://100.102.110.115:8444`. The `.ts.net` name works too.
4. The browser warns that the connection is not private (PortableTAK uses its own certificate authority). Choose **Advanced** → **Proceed** (Chrome) or **Show Details** → **visit this website** (Safari).
5. Sign in with the administrator password, the same as on Wi-Fi.

> If the page does not load: the Tailscale app must be **on** (check it shows *Connected*), the server must have internet, and the address must start with `https://` and end with `:8444`. On the server's own Wi-Fi you can also use `https://10.41.254.1:8444` (node).

### If remote access does not work

| Symptom | Fix |
|---|---|
| Node: "Could not start" after Enable | The node has no internet. Connect its Ethernet to a router or hotspot and try again. |
| "Could not start" after Enable on a computer | The computer has no internet, or a firewall blocks Tailscale. Check the internet connection and try again. |
| Phone stays red on cellular | Tailscale app off or signed in to a different account; the phone was set up before Step 1 (set it up again: ATAK QR Setup or a new package); the server side lost internet. |
| Works on cellular but not on the server's Wi-Fi | Normal for iTAK if only the Remote Package is imported: also import the normal iTAK Package. |

## 7. Drone video (Skydio X10D and other RTSP drones)

A drone can send live video to PortableTAK, and every TAK client sees it in its video list. This is the PortableTAK version of a Sit(x) Video Alias. Drone feeds need no sign-in to send or watch, so anyone who can reach the server (its Wi-Fi or mesh, or your Tailscale) can watch them.

### Connect the drone to PortableTAK (its position on the TAK map)

1. Dashboard → **Users** → create a user for the aircraft (e.g. `skydio1`) → **Device TLS Package**. Unzip it onto a USB-C stick. It holds the trust store, the client certificate, `secure.pref` and a **README.txt** with every value to type (the same layout as a Sit(x) TLS package). Keep it private: it contains the certificate passwords.
2. On the controller: Global Settings → Sharing → TAK/CoT → TAK Connections → Add a Connection. Server name (one word), server IP and port **8089** from the README. Advanced Options → SSL/TLS: the trust store (`truststore-root.p12`) and the client certificate (`skydio1.p12`), each with the password from the README.

### Step 1: add a feed (dashboard)

1. Dashboard → **Video Feeds** → type a feed name → **Add feed**. Use lowercase letters, numbers and dashes, for example `skydio1`. One name per aircraft.
2. The card then shows the exact drone settings, with this server's address. **Drone settings** shows them again later.

### Step 2: set up the Skydio X10D controller

1. Connect the controller's Wi-Fi to the same network as the server (the node's Wi-Fi, or the Wi-Fi the Windows/Mac computer uses).
2. Menu (three lines, top left) → **Sharing** → **RTSP** → **External**.
3. Stream name: the feed name (e.g. `skydio1`). IP address: the server's address shown in the card (node Wi-Fi: `10.41.254.1`). Port: **8554**.
4. **Start Stream**. The video starts once the aircraft takes off; the card shows **Live**.

### Step 3: watch it

ATAK / WinTAK: open **Video** and pick the feed by name. Or add it by hand: type RTSP, address `<server>:8554/skydio1`, and leave user name and password blank.

### Flying away from the server: Skydio on a phone hotspot

When the drone team is on cellular and the server is elsewhere, the controller joins an ATAK phone's hotspot and that phone forwards the drone's traffic to the server over its own Tailscale. Nothing is installed on the controller; the phone needs the free **Termux** app. Dashboard → **Video Feeds** → **Drone far from this server** has the one command to paste. Step by step: **PortableTAK Skydio X10D Guide** (Setup B).

### Channels

Channels (TAK groups) split who sees what. Dashboard → **Channels**: add, rename or delete channels, then pick a user and tick the channels they **send** into and **receive** from. Users in no channel use `__ANON__` (everyone without a channel). Phones pick up changes when they reconnect, and users can switch their channels on and off in ATAK/iTAK (TAK channels). `__ANON__`, ADS-B, AIS and Meshtastic are built in and cannot be renamed or deleted.

## 8. Linking servers (federation)

Federation shares positions, markers, alerts and chat between two TAK servers, so each team sees the other on its map. It works with another PortableTAK, OpenTAKServer with PortableTAK's federation add-on, and the official TAK Server (tak.gov). Dashboard → **Federation**.

**Data packages** an ATAK user sends to someone on the other server also cross between PortableTAK servers: the file moves over the federation link and the receiving phone downloads it from its own server. Not yet with an official TAK Server, and Data Sync missions are not shared.

### Two PortableTAK servers: one code

1. On the first server: **Invite a server**. Copy the code (or download the invite file) and send it to the other server's admin. It works once, for 24 hours.
2. On the second server: **Join a server**, paste the code, **Join**. Both servers trust each other, share with everyone, and the link turns green within a minute.

The servers need a network path to each other: the same network, or both drives' Remote Access signed in to the same Tailscale account.

### An official TAK Server

1. **Connect to a TAK Server**: name, address and federation port (9001), then **Check**. PortableTAK reads the TAK Server's certificate authority itself and shows its fingerprint; have the TAK Server admin confirm it if you can, then **Trust and connect**.
2. Send the TAK Server admin the sheet that appears (it is also under Advanced): this server's CA certificate to upload under *Federation* > *Manage Federate Certificate Authorities*, then, once PortableTAK has connected, the groups to allow for its federate (users without a group are `__ANON__`).

### Choosing what flows

Each linked server shows this server, two arrows and the other server. Tap → to send to it or stop sending, ← to receive from it or stop receiving (green = on); the line underneath says which way data flows. **Pause** stops a link without deleting it. **Channels...** on a link chooses which channels it sends and which channels the other server's data arrives in (official TAK Servers match channels by name).

### Good to know

- A server that connects to yours without an invite appears as *not sharing yet*: press **Allow sharing** only if you trust it.
- Each link is locked to the other server's certificate. If that server is rebuilt, use **Forget its certificate** (or remove the link and invite again).
- Moving the drive to another node or computer keeps its links: the federation certificate belongs to the drive.

## 9. Backups, moving and erasing

- **Backups: Backup / Restore** creates checked backups on the drive. Download important ones and keep them elsewhere.
- **Moving the drive** to another node or computer: stop or safely remove it, plug it in elsewhere. It adapts by itself.
- **Erase Drive** (bottom of the dashboard): returns the drive to brand new. It deletes all users, certificates, settings, remote access, data and on-drive backups. It asks for the administrator password and the word ERASE, and cannot be undone. Set the drive up again with PortableTAK Setup afterwards.

## 10. Troubleshooting

| Problem | What to do |
|---|---|
| Phone says "host unreachable" or never connects | Check the phone is on the node's Wi-Fi (or the same network as the Windows computer). Use the right address (node Wi-Fi: 10.41.254.1). On Android, set the Wi-Fi to DHCP (not static); if it still fails, forget and rejoin the network. Over cellular, the Tailscale app must be on. |
| Connection red only on cellular | Remote Access must be enabled and finished, the node/computer needs internet, and the phone's Tailscale app must be connected. |
| Another device shows grey or disappears | Make sure each device has a different callsign. Grey can also mean that device lost its connection; it returns when it reconnects. |
| iPhone sees only one server | Expected: iTAK reads one server per package. Import the **iTAK Remote Package** as well. |
| ATAK says "registration failed" or "TAK Server profile request failed" after QR setup | The phone scanned a QR made before an update, or the server's address changed since. Delete the server in ATAK, then scan a fresh **ATAK QR Setup**. If the server still turns green, it is harmless. |
| iTAK QR: "There was an error connecting to the server" | The iPhone does not fully trust this drive yet. Installing the **iPhone Trust Profile** is not enough: also go to Settings → General → About → **Certificate Trust Settings** and switch **on** this server's certificate. Each drive has its own profile (one made for another drive does not count). The **iTAK Package** works without this. |
| "Invalid credentials" / QR enrollment fails | Use the user's own name and password, not the administrator. ATAK must trust the server first (import the ATAK Trust Package once). Generate a new QR (each works 3 times). |
| Certificate warning in the browser | Normal: the server uses its own certificate authority. Choose Advanced → Continue. To remove it, install the drive's trust certificate on that device. |
| Dashboard sign-in refused after several tries | Wait a few minutes (sign-in is rate-limited per device), then use the administrator password chosen at setup. |
| Node: server does not start after plugging in | Wait 5 minutes. Re-seat the drive firmly in a blue USB 3 port. Power-cycle the node; PortableTAK starts on boot. |
| Windows: "No PortableTAK drive found" | Plug the drive in and wait until the PORTABLETAK drive appears in File Explorer, then start again. |
| Windows: asks to restart | WSL was just installed. Restart, then start PortableTAK again. |
| Windows: phones cannot connect | Phones must be on the same network as the computer. Changed Wi-Fi since starting? Use **Stop PortableTAK** and start it again so it uses the new address, then set phones up again (QR or package). Windows 11 22H2+ is required. |
| ATAK: "The TAK Server's identity could not be verified" after QR setup | ATAK no longer trusts the drive (deleting servers in ATAK can remove the trust). Import the **ATAK Trust Package (for QR)** again from Client Setup, |

| | |
|---|---|
| | then scan a fresh QR. Or import the **ATAK Package** for that user, which carries everything ATAK needs. |
| Federation link stays red | "Not connected: certificate" after the other server was rebuilt: **Forget its certificate**, or remove and invite again. "Connection refused" or unreachable: the servers need a network path (same network, or the same Tailscale account). A TAK Server peer: its admin must upload this server's CA and set the federate's groups. |
| Drone feed never shows Live | The controller must be on the server's network, with the IP address and port **8554** from the Video Feeds card and exactly the same stream name. Video only starts once the aircraft is flying. |
| Windows: drive vanished from File Explorer | Normal while PortableTAK runs. Use **Stop PortableTAK** to get it back. |
| Windows offers to format the drive, or macOS to initialize it | Always Cancel / Ignore. The Linux partition holds the server. |
| "More than one PortableTAK drive is attached" / server does not start with two drives plugged in | Use one PortableTAK drive per node or computer at a time. Unplug the other drive and start again (node: power-cycle or re-plug). |
| The drive was unplugged without Safe Remove / Stop PortableTAK (or power was lost) | Usually no harm: plug it back in and start PortableTAK. It checks and repairs the drive automatically and cleans up the previous session. Next time use **Safe Remove** (dashboard) or **Stop PortableTAK** (desktop) first. |
| Map icons missing | They download automatically the first time the server runs with internet. |
| Forgot the administrator password | It cannot be recovered. Use a backup made elsewhere, or erase and set up the drive again. |
| Node: a new node does not start PortableTAK | The node needs the PortableTAK loader once (field extension or `portabletak-loader-install.sh`). On the node, `portabletak-loader status` shows what happened; a refused drive means its package is not signed with the PortableTAK release key (make the drive again with the official Setup). |

### Logs

**Windows:** `%LOCALAPPDATA%\PortableTAK\portabletak.log`. **Node** (administrators, over SSH): `/usr/local/portabletak/bin/portabletak-health` and the logs under `/mnt/asgard/opentak/data/logs`.

## 11. Security notes

- The dashboard requires sign-in. It is reachable on the TAK network at port 8444 (HTTPS), and with remote access also from devices signed in to your Tailscale account (nobody else can reach it). It is protected against cross-site requests and password guessing. Choose a strong administrator password for real use.
- First-time setup can only be done on the computer that runs the drive, never from a phone.
- The drive contains the server's private keys. Keep it physically secure. Erase drives before giving them away.